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Printed Pages – 7

324313 (24)

BE (3rd Semester)

Examination, April-May, 2013

Branch : Electrical

SOLID STATE DEVICES

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part 'a' is compulsory and carry two marks.

(ii) Attempt any two parts from remaining of each unit. All questions carry equal marks.

Q. 1. (a) Draw a circuit model of a diode & V-I characteristics of an ideal diode. 2

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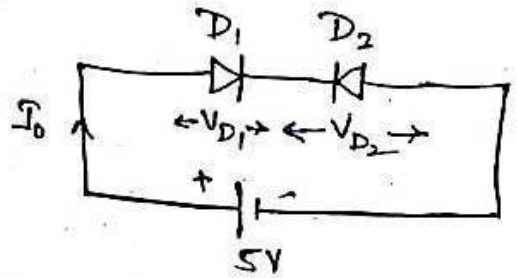
(b) Sketch V-I characteristics of a diode and explain. 7

(c) Derive the expression for the transition capacitance (C_T). 7

(d) Two p-n germanium diodes are connected in series opposing. A 5V battery is impressed upon this series arrangement. Find the voltage across each at room temperature. Assume that the magnitude of zener voltage is greater than 5V. 7



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Q. 2. (a) What is Peak Inverse Voltage (PIV) ? 2

(b) Write down the comparison of full wave and half wave circuit. 7

(c) Draw and explain the full wave bridge rectifier circuit. What are the advantages of bridge rectifier circuit. 7

(d) Prove that maximum rectification efficiency of full wave & half wave rectifier is 81.2% and 40.6% respectively. 7

Q. 3. (a) What is dc load line of a transistor ? 2

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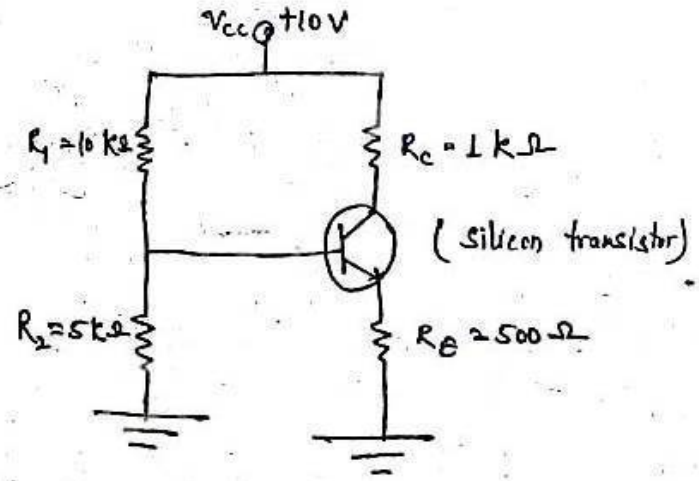
(b) For the circuit shown below : 7

β = 100 for the silicon transistor. Calculate V_{CE} & I_C.



(c) For the circuit given below, determine I_E, V_E and V_{CE}. 7

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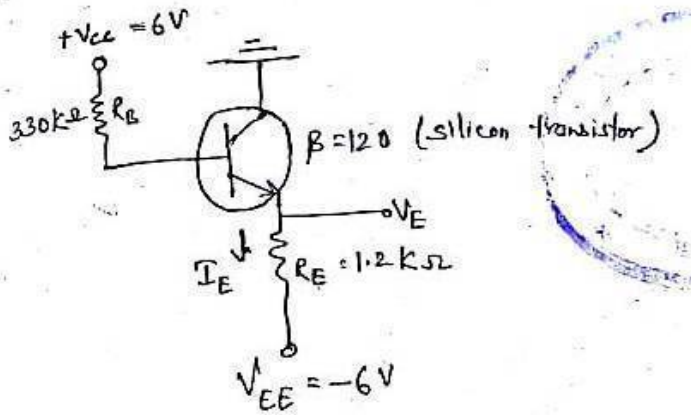

(d) Explain any one method of biasing a single stage BJT amplifier. 7

Q. 4. (a) What is amplification factor μ for JFET. 2

(b) Explain the construction, operation and characteristics of depletion type MOSFET. 7

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(c) Prove that, for the JFET : 7

$$g_m = -\frac{2I_{DSS}}{V_P}\left(1 - \frac{V_{GS}}{V_P}\right)$$

(d) Explain the drain characteristics and transfer characteristics of N-channel JFET. 7

Q. 5. (a) What do you mean by positive feedback amplifiers ? 2

(b) Explain RC phase oscillator with diagram. 7

(c) Explain construction & working of Hartley's oscillator. 7

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(d) What are the effects of negative feedback amplifier ? Explain. 7

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